Hack the Fork Paris 2026: the 5 ideas

Track 3, protein sovereignty: how AI can help France grow more pulses for human food. Plain-language summary of the 5 audited ideas, with their scores under the jury criteria. 30 October 2026.

The recommendation in one line

Archive: earlier comparison of ideas 1 to 5. The team later chose SoyMate (phone-photo quality control of pulses); see CONTEXT.md. **Go with idea 1 (pea protein)**, adding the explanation with twin seasons (idea 2) and, if time allows, the protein radar by basin. **Alternative: idea 3 (bruchids)**, the best impact story but with more fragile data. **Plan B: ideas 4 and 2 together** (early yield forecast explained with twin seasons). **Discard idea 5** (drone). None exceeds 6.4: the score goes up with mentor validation and real results.

How the jury scores

Each juror gives a score from 1 to 10 per criterion, weighted as follows. Impact weighs more than everything else, and AI has to be essential: if you remove it from the project and it does not lose much, the score drops.

Criterion	Weight	What it looks at, in plain terms
Impact and sustainability	30%	Does it really help a more sustainable food system? Are the benefits backed by evidence?
Scientific rigor	20%	Are the data and sources serious? Does the team acknowledge the limits of its solution?
Essential AI	20%	Is AI the heart of the project or decoration? Does the team explain what the model does?
Economic viability	20%	Is there a concrete customer who would pay? Is there a realistic path for them to adopt it?
Execution and pitch	10%	Does the demo work live? Does the team answer questions well?

Rubric: where each idea is strong

Scores estimated by us from the audit of each idea, before training any model (the rules forbid it before the event). Green = strong point, red = weak point.

Idea	Impact 30%	Rigor 20%	Essential AI 20%	Economics 20%	Execution 10%	Total
1 · Pea protein for industry	6	7.5	6.5	5.5	7	**6.4**
2 · Twin season (climate analogues)	6	7	6	5	7	**6.1**
3 · Bruchid risk in lentil and faba bean	6	7	5.5	5	7	**6.0**
4 · Early département-level yield forecast	5.5	7.5	5.5	4.5	7.5	**5.9**
5 · Trial yield from drone imagery	4	7	7	4	6	**5.4**

- **Best on impact:** 1, 2 and 3 tie at 6; 3 has the most direct story (damaged grain = not fit for human food).
- **Best on rigor:** 1 and 4 (7.5): public, audited data and year-by-year validation.
- **Best on AI:** 5 (vision), but it loses on everything else; among the viable ones, 1 (6.5).
- **Best on economics:** 1 (5.5): concentrated customer with a real pain (tariffs on Chinese pea protein).
- **Best on execution:** 4 (7.5): data and pipeline proven end to end.

Idea 1: Pea protein for industry

Score: 6.4 / 10 · Verdict: **Main (adding the explanation from 2 and, optionally, the basin radar)**

Will this pea have enough protein to make ingredients?

Pea protein factories (such as Roquette or Cosucra) contract peas months before harvest, without knowing how much protein they will carry. Our tool looks at the season's weather and estimates the probability that each region's peas reach a useful protein level (for example 23%), and updates it at flowering and before harvest.

Who for	Supply manager at a pea protein factory, or the cooperative that sources for it. Decides in September (winter pea) and in December-January (spring pea).
What the AI does	Model that learns from 12 years of trials how the weather moves protein and yield, and gives a probability with its margin of error.
With what data	4,450 pea trials from INRAE, Terres Inovia and GEVES (2002-2024, published August 2026) + daily SAFRAN weather from Météo-France.
Hook	From 26 September 2026 the EU charges up to 67% duties on Chinese pea protein: European factories need local peas.

Why it can win	Why it can lose
+ The cleanest and newest data of everything we audited: we found few teams using it so far. + Very strong news hook (antidumping duties). + Clear demo: rerun 2018 (bad year) and 2023 (good year) without the model having seen them.	– Contracts today do not pay for protein: confirm with mentors that knowing it earlier is useful to the factory. – Only 12 years of protein data: the model has to beat simple rules. – What decides whether a pea is fit for human food is split grains, not protein.

Optional: protein radar by basin (what used to be idea 6). An extra screen that multiplies pea area (Agreste publishes it every month by département) by expected yield per expected protein, and shows the factory how many tonnes of protein it will have available in its purchasing basin, with its error range. Takes 2 to 3 hours if idea 1 already works. Adds a figure that is easy to grasp for the pitch and an extra validation with 35 basin protein averages from Terres Inovia. Careful: at national level protein explains only 2% of the variation in tonnage (area 68%, yield 30%), so show it by basin and not as a national radar. Do it only after Saturday's checkpoint.

Idea 2: Twin season (climate analogues)

Score: 6.1 / 10 · Verdict: **Very good as a layer on another idea**

Which past season does this one resemble, and what harvest did that year give?

Agronomists already reason this way: "this year looks like 2016". With the weather up to today, the tool searches for the most similar past seasons in that département and shows what pea yields they gave. From that it gives an estimate with a range and explains why.

Who for	Collection manager at a cooperative that sells peas to the food industry; decides volumes and forward sales in May-June.
What the AI does	Case-based reasoning: a similarity measure learned from the data, which finds twin seasons and explains the estimate.
With what data	Agreste département yields 2000-2025 (26 pea seasons in 43 départements) + SAFRAN weather since 1958.
Hook	"This year looks like 2016" (the worst recent harvest), said with numbers and before anyone else.

Why it can win	Why it can lose
+ Very easy to explain to the jury and to a farmer.	– It may not beat the average of the last 5 years.
+ We found more history than expected: 26 seasons, not 15.	– The trend has to be corrected: peas yield less and less (from 45 to 30 q/ha).
+ Combines naturally with ideas 4 and 1 as an explanatory layer.	– Still need to confirm that a cooperative makes any decision in May-June based on this.

Idea 3: Bruchid risk in lentil and faba bean

Score: 6.0 / 10 · Verdict: **Alternative with the strongest story**

Which lentil fields will be most damaged by the bruchid?

The bruchid is an insect that lays its eggs on the pod, and the larva eats the grain. A perforated grain cannot be sold for human food. In the 2025 national quality survey, lentil lots had on average 24.4 % bruchid-damaged grains, and since July 2024 no insecticide is authorised in the field against bruchids. The tool estimates, at flowering, how much will be lost at sorting in each field, and shows on a map where to sow lentil the following year.

Who for	Lentil filière manager at a cooperative (for example in the Aube): plans sorting, lots and sowing.
What the AI does	Model that combines landscape (how much lentil was grown nearby the year before), weather and crop to estimate the damage with its uncertainty.
With what data	INRAE bruchid dataset (100 fields, 2020-2022) + RPG parcel map to compute the landscape of any field in France.
Hook	24.4 % of grains bruchid-damaged on average in the 2025 lots (the highest value in the survey) and no insecticide in the field: we found no tool that predicts the damage per field.

Why it can win	Why it can lose
+ It is the idea with the most direct impact on human food and on the farmer's income.	– Only 100 fields and 3 years: hard to beat the average by region and year.
+ No competition: we found no French tool that predicts damage per field.	– Need to confirm that "insects" in the exclusions refers to insects as food.
+ The landscape can be recomputed for any plot in France with the RPG.	– Predicting the damage does not reduce it on its own: the benefit in tonnes is indirect.

Idea 4: Early département-level yield forecast

Score: 5.9 / 10 · Verdict: **Solid plan B**

How much pea will my département harvest this year?

With the weather up to mid-June, the tool estimates the pea and faba bean yield of each département, with an error range, before the State publishes its first estimate (1 July). It helps the cooperative know how much grain it will receive and how much it can sell.

Who for	Collection manager at a cooperative: decides storage, forward sales and contracts.
What the AI does	Forecasting model with weather weighted by where the plots actually are, with a confidence interval.
With what data	Agreste 2010-2025 (38 usable pea départements) + SAFRAN + RPG parcels to weight the weather.
Hook	Knowing two or three weeks before the official estimate, and with an interval.

Why it can win	Why it can lose
+ Data and pipeline proven end to end: low technical risk. + A lot of rigor possible: year-by-year validation, data errors detected. + Visual demo with a département map.	– Agreste already publishes département estimates from 1 July, for free. – Part of the pea and faba bean goes to animal feed, which is excluded: the human-food focus has to be justified. – Lentil and chickpea cannot be validated (imputed data).

Idea 5: Trial yield from drone imagery

Score: 5.4 / 10 · Verdict: **Discard**

Which pea and chickpea varieties will yield most, judging from drone photos?

With drone photos taken before harvest, estimate the yield of each trial plot so that a plant breeder can drop the bad varieties earlier. The data comes from trials in the United States.

Who for	Plant breeder or variety trial network.
What the AI does	Computer vision on multispectral images + prediction with an interval.
With what data	WSU/USDA dataset (Washington and Idaho, 2017-2019): drone images and yields per plot.
Hook	None strong for France.

Why it can win	Why it can lose
+ AI is clearly essential (computer vision). + Very visual demo.	– US data, a user far from the farmer, and Hiphen already offers this. – Uncalibrated images, misaligned between dates: 2 to 4 hours just to locate the plots. – There are already papers using these same data.

What to confirm on Friday before 23:00

- With a mentor: would a pea protein factory find it useful in September to know how much protein it will receive? And prepare the answer to "protein explains only 2% of the tonnage" (decides 1)
- With the organizers: that the "insects" exclusion refers to insects as food, not pests (decides 3).
- With the organizers: that working with pea and faba bean is eligible even though part of it goes to animal feed (affects 2 and 4).
- With a cooperative mentor: is any volume decision made in May-June? (decides 2)

Full detail for each idea (verified data, hypotheses, hour-by-hour plan, pitch, risks and sources): research/planes/ in the project repository. Dataset map: research/datasets.md.